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Course/Section: CPE221/CPE31S1	Date Submitted: 29/01/2026
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Procedure:

1. Note: You are required to create a document report of the steps you will do for this exam. All screenshots should be labeled and explained properly. LABELED AND EXPLAIN EACH CODE (PLAYBOOK) No explanation = Minus Points
2. Create a repository in your GitHub account and label it as Surname_PrelimExam
3. Clone your new repository in your CN.
4. In your CN, create an inventory file and ansible.cfg files.
5. Create an Ansible playbook that does the following with an input of a config.yaml file for both Manage Nodes
 - Installs the latest python3 and pip3
 - use pip3 as default pip
 - use python3 as default python
 - Install Java open-jdk
 - Install MariaDB as well as starting the server, create a database and a table using mariaDB and input one record into a table USING ANSIBLE ONLY
 - Create Motd containing the text defined by a variable defined in config.yaml file and if there is no variable input the default motd is "Ansible Managed node by (your user name)"
 - Create a user with a variable defined in config.yaml

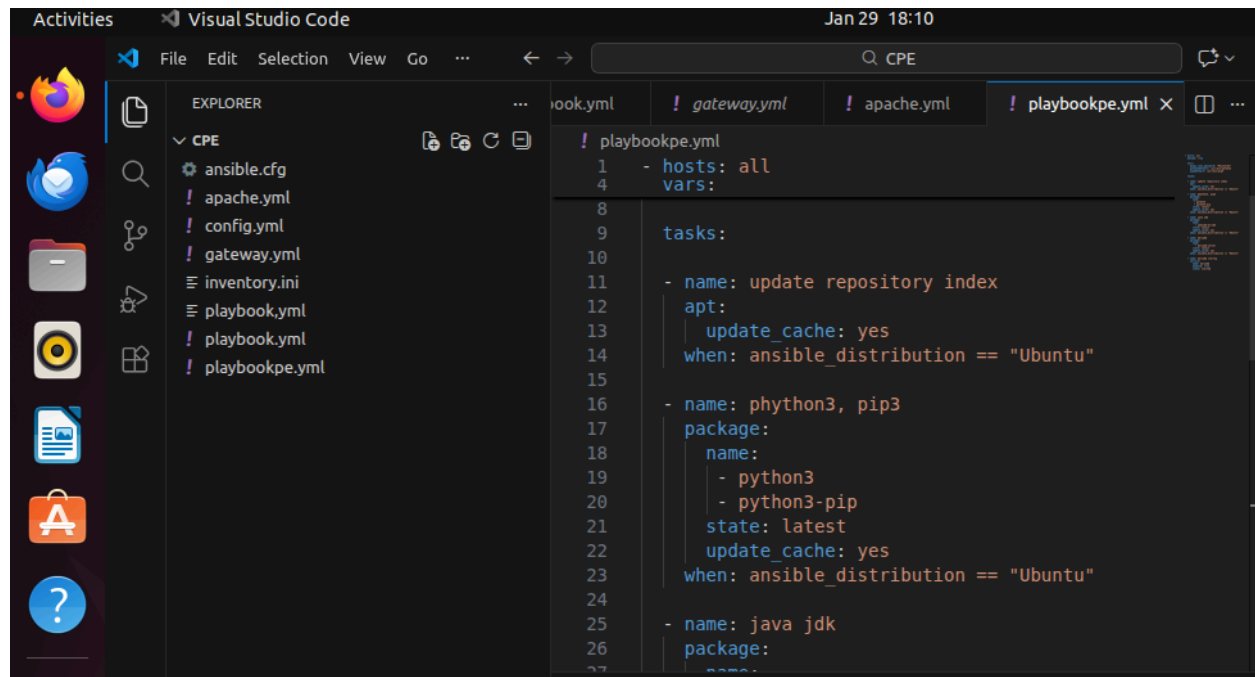
5. **PUSH and COMMIT your PrelimExam in your GitHub repo**

6. Your document report should be submitted here.

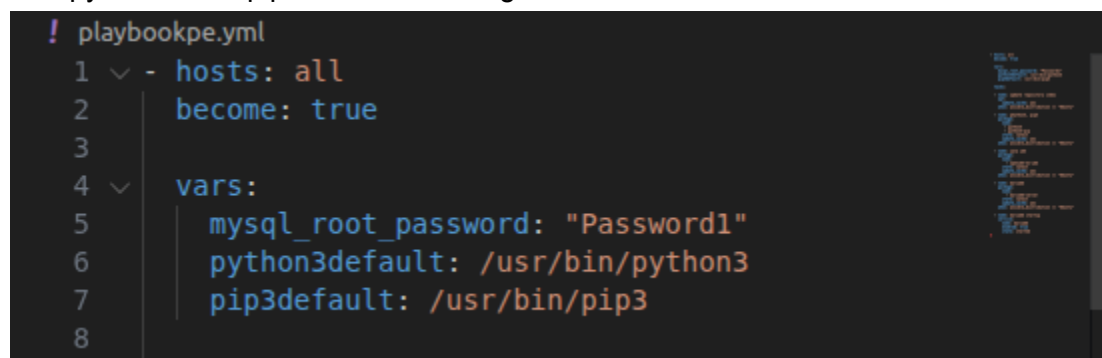
7. For your prelim exam to be counted, please paste your repository link here. (Failure to submit will result in ZERO)

8. NO USE OF EXTERNAL WEBSITES SUCH AS REDDIT, CHATGPT, GITHUB, GEMINI, AND CLAUDE. FAILURE TO COMPLY WITH RESULT IN ZERO. AI OVERVIEW SEARCH WILL ALSO RESULT IN A FAILED GRADE. Always use the "-ai" in the search bar, only use Google Search Engine.

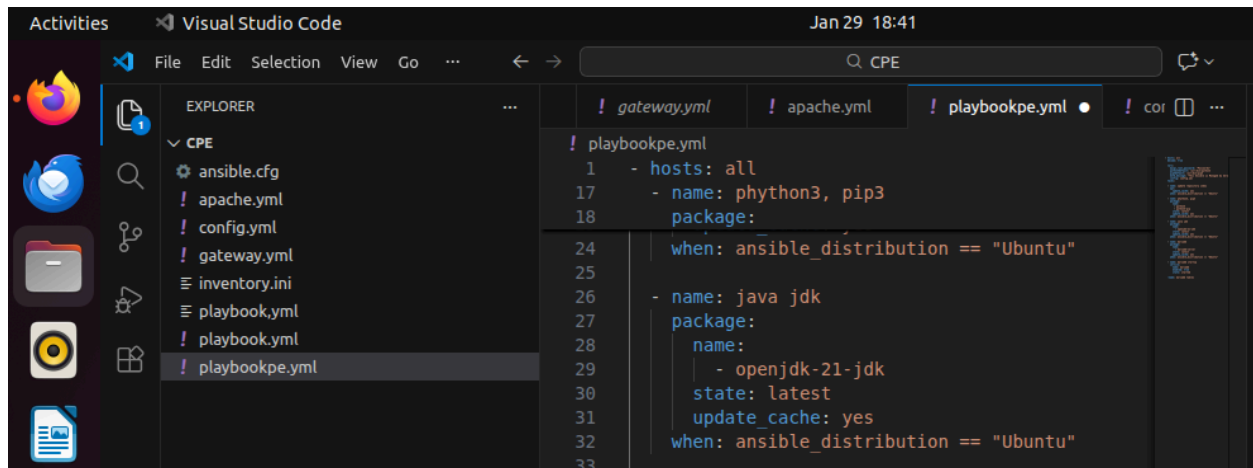
Installing python 3 and pip3, package, name: python3 and python3-pip state is set to latest.



I set python 3 and pip3 as default using the code below



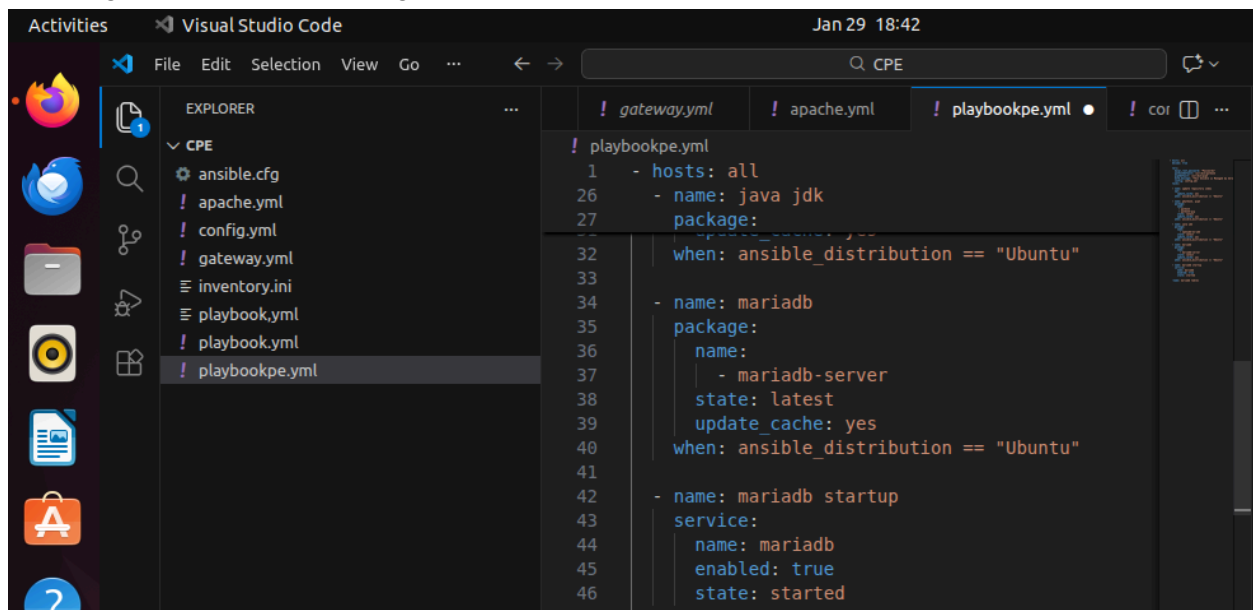
Installing java jdk using the package openjdk-21-jdk



The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying a file tree for a project named 'CPE'. The file tree includes files like 'ansible.cfg', 'apache.yml', 'config.yml', 'gateway.yml', 'inventory.ini', 'playbook.yml', and 'playbookpe.yml'. The 'playbookpe.yml' file is selected and its content is displayed in the main editor. The playbook is written in YAML and includes a task to install the 'java jdk' package using 'openjdk-21-jdk' as the package name. The task is conditional on the 'ansible_distribution' being 'Ubuntu'.

```
! playbookpe.yml
1 - hosts: all
17 - name: python3, pip3
18   package:
24     when: ansible_distribution == "Ubuntu"
25
26 - name: java jdk
27   package:
28     name:
29       - openjdk-21-jdk
30     state: latest
31     update_cache: yes
32   when: ansible_distribution == "Ubuntu"
33
```

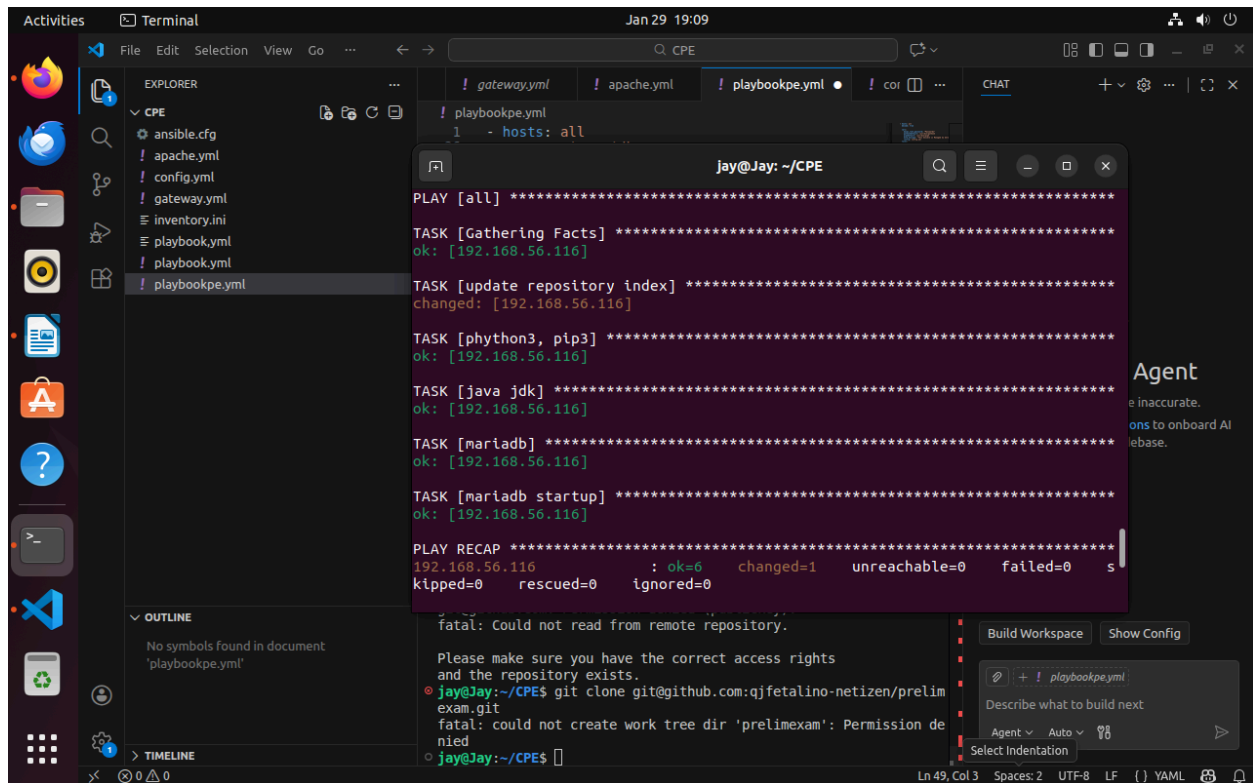
Installing maria DB and starting it up



The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying a file tree for a project named 'CPE'. The file tree includes files like 'ansible.cfg', 'apache.yml', 'config.yml', 'gateway.yml', 'inventory.ini', 'playbook.yml', and 'playbookpe.yml'. The 'playbookpe.yml' file is selected and its content is displayed in the main editor. The playbook is written in YAML and includes a task to install the 'mariadb' package using 'mariadb-server' as the package name. The task is conditional on the 'ansible_distribution' being 'Ubuntu'. Additionally, there is a task to start the 'mariadb' service.

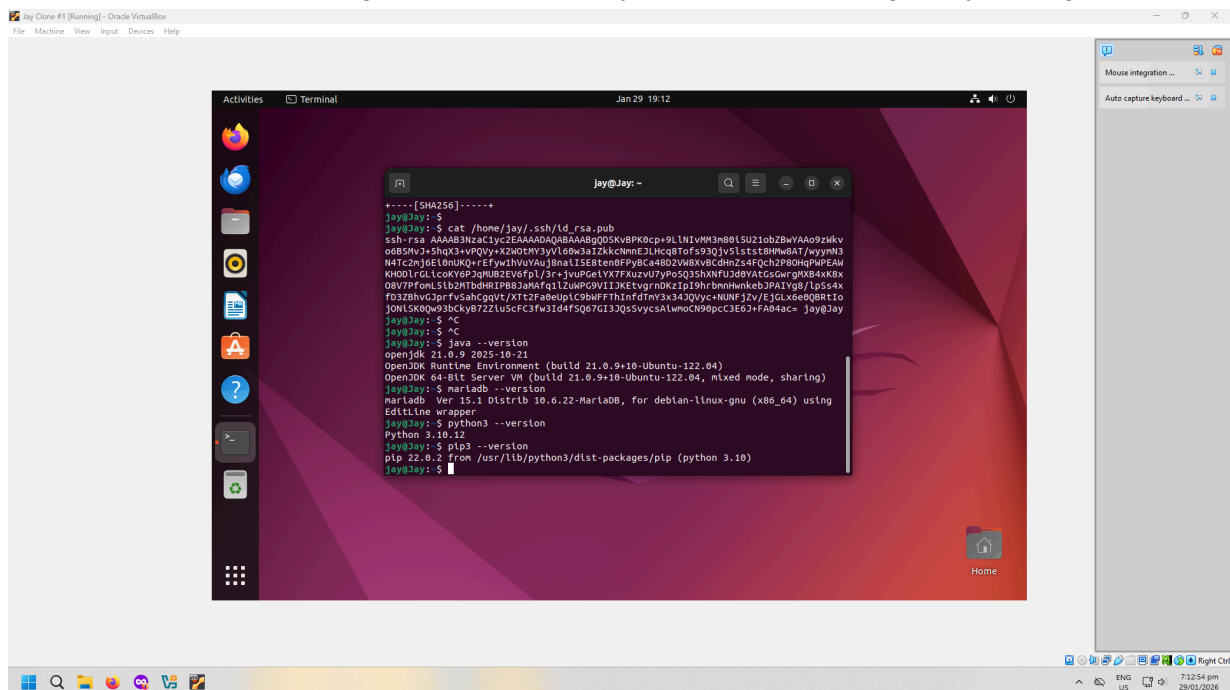
```
! playbookpe.yml
1 - hosts: all
26 - name: java jdk
27   package:
32     when: ansible_distribution == "Ubuntu"
33
34 - name: mariadb
35   package:
36     name:
37       - mariadb-server
38     state: latest
39     update_cache: yes
40   when: ansible_distribution == "Ubuntu"
41
42 - name: mariadb startup
43   service:
44     name: mariadb
45     enabled: true
46     state: started
47
```

Screenshot of playbookpe.yml showing that its been working so far



```
PLAY [all] *****
TASK [Gathering Facts] *****
ok: [192.168.56.116]
TASK [update repository index] *****
changed: [192.168.56.116]
TASK [python3, pip3] *****
ok: [192.168.56.116]
TASK [java jdk] *****
ok: [192.168.56.116]
TASK [mariadb] *****
ok: [192.168.56.116]
TASK [mariadb startup] *****
ok: [192.168.56.116]
PLAY RECAP *****
192.168.56.116 : ok=6 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
fatal: Could not read from remote repository.
Please make sure you have the correct access rights
and the repository exists.
jay@Jay:~/CPE$ git clone git@github.com:qjfetalino-netizen/prelimexam.git
fatal: could not create work tree dir 'prelimexam': Permission denied
jay@Jay:~/CPE$
```

Screenshot of proof showing that all apps in playbook that its working in my managed nodes



```
jay@Jay: ~
+----[SHA256]-----+
jay@Jay: $ cat /home/jay/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQGDQKvBPK0cp+9LLN1vM3m01SU21ob2BwYAA09zKv
o6B5Mv3+ShqX3+VpQVv+X2W0tMv3yVl6w3a1ZkcnmE2LHcBtOfs930jv51ststBHWBAT/wyyn3
NATc2nJ0e10nUQ+EfYw1hVvUj8nallSEBten0PyBCa4802VMBXv8CdhZs4Fqch2P80HqKPEAM
NH001GLC0KvG2pQmUB2E0fzJr3v0P0EYAT7Kuz4U7P0S03ShNvUJ20YATC0wgnBvK8K
0BV7PF0mL5lb2MtdBHRIPB8JaMaq11ZuMPG0V113KETvgrnDk2Ipi9hrbnnHnkeb3PA1Yg8/lp554X
fD3Z8hvG3prfV5ahCqvT/XT2fabeUplC9BwFFThInfTmV3x343Qvyc+NUNFJ2V/EJLx0e0Q8RTio
JONLSK0w93bckYb722Lu5CFC3fW31d4f5Q67G13Q55vycsAlwmoCh9pcc3E0J+FA04ac= Jay@Jay
jay@Jay: $ ^C
jay@Jay: $ ^C
jay@Jay: $ java --version
openjdk 21.0.9 2025-10-21
OpenJDK Runtime Environment (build 21.0.9+10-Ubuntu-122.04)
OpenJDK 64-Bit Server VM (build 21.0.9+10-Ubuntu-122.04, mixed mode, sharing)
jay@Jay: $ mariadb --version
mariadb Ver 15.1 Distrib 10.6.22-MariaDB, for debian-linux-gnu (x86_64) using
Editline wrapper
jay@Jay: $ python3 --version
Python 3.10.12
jay@Jay: $ pip3 --version
pip 22.0.2 from /usr/lib/python3/dist-packages/pip (python 3.10)
jay@Jay: $
```

